CAPSTONE · AI FOR GOOD · WEEK 4

How an offshore wind farm makes, *or loses, money*

The wind is free. The money is not. Tonight: why a change in interest rates can cancel a wind farm that the wind would have paid for.

Dr. Jan Erik Meidell · Wednesday 4 November 2026 · online · 18:15–20:00 · the Week 4 page open in a second tab (the calculator)

TONIGHT

One hour forty-five, two exercises

18:18	A guess in the chat
18:21	What a wind farm costs: to build, to run, per MWh
18:35	The cost of money: interest rates and the cost per MWh
18:45	Exercise 1 · three shocks on the calculator
18:58	How a wind farm gets paid: contracts for difference, auctions, PPAs
19:15	Break · 5 minutes
19:20	Ørsted and Eni: two ways to pay for the transition
19:38	Exercise 2 · bid or walk away
19:52	Into the report · close

A GUESS IN THE CHAT · A, B, C OR D

How much of the cost of each MWh is the cost of the money?

1. A · about 10%
2. B · about 30%
3. C · about 50%
4. D · about two thirds

D. For a UK offshore farm at a 6.2% cost of capital, paying back the money spent upfront is about two thirds of the cost of every MWh. The rest is running the farm. We will check it by hand in five minutes.

THE SHAPE OF THE BUSINESS

A wind farm is a loan with blades

Year 0

Almost all the money is spent before the first MWh is sold: turbines, foundations, cables, a substation, installation ships.

Years 1–35

The fuel is free and running costs are modest. The farm earns its money back slowly, MWh by MWh, over decades.

So the price of a wind MWh is mostly the price of money borrowed or invested years earlier. A gas plant is the opposite: cheap to build, and the fuel is paid as you go.

TO BUILD · UK FIXED-BOTTOM OFFSHORE, 2023 PRICES

About $5.3 million per MW

WHERE THE MONEY GOES	SHARE OF CAPEX
Turbines (supply 35%, installation 3%)	**38%**
Foundations (supply 16%, installation 8%)	**24%**
Offshore substation	**12%**
Export cable to shore (5% + 2%) · cables between turbines (3% + 1%)	**11%**
Other: development, insurance in construction, contingency	**15%**

Capex $5,330/kW (range $4,460–5,950), projects commissioning 2028–32: Arup for the UK energy department (DESNZ), July 2025, in pounds (£3,976/kW), converted at $1.34 per £ (4 November 2026); every dollar figure in this deck uses that rate. A peer review says the sample leans high. A real check: Ørsted puts Hornsea 3 (≈ 2.9 GW) at DKK 70–75 bn, about DKK 25 m per MW.

TO RUN

About $186,000 per MW, every year

$62.4k

Operations and maintenance: technicians, vessels, spare parts

$111.9k

Grid connection and use-of-system charges: more than half of the running cost

$11.5k

Insurance

And it produces: 1 MW × 8,760 hours × a 50.5% load factor = **4,424 MWh a year**, for 35 years.

Arup / DESNZ, July 2025, 2023 prices. The load factor range is 46–56%.

BY HAND · FROM COSTS TO A PRICE PER MWH

What one MWh must earn

STEP	CALCULATION	PER MW
Spread $5,330k over 35 years at 6.2% (like a mortgage)	$5,330k × 0.0706	**$376k a year**
Add the running cost	$376k + $186k	**$562k a year**
Divide by what it produces	$562k ÷ 4,424 MWh	**≈ $127 per MWh**
The capital part of it	$376k ÷ 4,424 MWh	**$85 = 67%**

0.0706 is the annuity factor: $0.062 \div (1 - 1.062^{-35})$. Arup's full model, with construction time and decommissioning, gives $118.7/MWh (£88.5). The simple version is close enough to reason with.

NAME IT

The levelised cost of electricity (LCOE)

LCOE = (capex spread over the life at the cost of capital + yearly running cost) ÷ MWh produced per year

- **Capex**: what it costs to build, per MW.
- **Cost of capital**: the return lenders and shareholders demand, often written WACC (weighted average cost of capital).
- **MWh per year**: capacity × 8,760 hours × load factor.

You met LCOE in Week 2 and fitted its fall in Week 3. Tonight is what is inside it.

THE COST OF MONEY

Same farm, same wind, different interest rate

COST OF CAPITAL	4.2%	6.2%	8.2%	10.2%
LCOE ($/MWh)	**108**	**127**	**148**	**169**
Change against 6.2%	**–15%**	**—**	**+16%**	**+33%**

Two percentage points on the cost of capital move the cost of every MWh by about 16%, and the turbines have not changed at all.

Simple model from slide 7. A peer-reviewed study finds the same order of magnitude: a 3.3-point higher cost of capital in Germany means 26% higher LCOE (iScience, 2023).

WHAT ACTUALLY HAPPENED

2022–2023: the price of money jumped

The central banks

ECB deposit rate: **–0.50%** until July 2022 → **4.00%** from September 2023.
US Federal Reserve: near zero in early 2022 → **5.25–5.50%** in July 2023, 525 basis points in total.

One company's arithmetic

Ørsted, fourth quarter of 2024: a **0.75-point** rise in the cost of capital it applied to its US portfolio led to a **DKK 4.3 bn** impairment. Three quarters of a percentage point, four billion crowns.

Projects priced in 2019–21 with cheap money were built with expensive money.

EXERCISE 1 · 11 MINUTES · BREAKOUT ROOMS · THE CALCULATOR ON THE WEEK 4 PAGE

Three shocks: which one hurts most?

1. **Predict first**, in the canvas box "Three shocks": which of the three below raises the cost per MWh most?
2. Start from the default farm ($5,330/kW · $186k · 50.5% · 6.2% · 35 years). Apply each shock **alone**, write the new LCOE:
   A · cost of capital **+2 points** · B · capex **+15%** · C · load factor **–5 points**
3. Then all three together. Was your prediction right?

Back in the main room at 18:56. One number per team.

READ-OUT

Rates, capex and wind, compared

Alone: rates +16%, capex +10%, load factor +11%. Together: about +43%, from $127 to $181 per MWh. 2022–23 delivered the first two at once, which is why projects with a fixed price agreed before 2022 stopped making sense.

The point for your scenarios: interest rates are a driver of wind's cost, independent of EU climate policy. That is what makes it a usable axis next to climate policy.

HOW THE FARM GETS PAID

Four ways to sell the power

Contract for difference (CfD)

The state guarantees a fixed price (the strike price) for 15–20 years. The most common model for new European offshore wind.

Merchant

Sell at whatever the market pays each hour. All the upside, all the risk, including the falling capture price (Week 5).

Corporate PPA

A long contract with a big buyer: Ørsted–Covestro, 10 years, 100 MW from Borkum Riffgrund 3; EnBW–Google, 15 years from He Dreiht.

Pay for the seabed

The developer pays the state for the right to build: Germany's June 2023 auction raised about €12.6 bn, with oil majors among the winners, paying €1.56–2.07 m per MW.

BY HAND · A TWO-SIDED CFD

The state tops up, or takes back

STRIKE PRICE $122.30/MWH	MARKET PRICE	THE FARM RECEIVES	THE STATE
A windless winter evening	**$160**	**$122.30**	**takes back $37.70**
An average hour	**$95**	**$122.30**	**pays $27.30**
A windy night	**$40**	**$122.30**	**pays $82.30**

The farm's revenue becomes predictable, so lenders accept a lower return. A CfD lowers the cost of capital, which (slide 9) lowers the cost of the power.

UK AUCTIONS · CONTRACTS FOR DIFFERENCE

Set the price too low and nobody builds

ROUND	OFFSHORE WIND RESULT	IN 2012 PRICES
AR5, September 2023	**No bids.** The ceiling was not raised for inflation and higher interest rates.	**ceiling £44 (≈ $59)**
AR6, September 2024	3.36 GW (Hornsea 4, East Anglia 2)	**£58.87 (≈ $79)**
AR7, January 2026	8.2 GW at £91.20 = **$122.30** (2024 prices), now with **20-year** contracts	**£65.45 (≈ $88)**

AR5 is slide 9 in real life: the government's price was set for cheap money, and the money was no longer cheap.

UK Department for Energy Security and Net Zero, allocation results. Wood Mackenzie estimates the 20-year term let bidders go about $13/MWh lower. The UK auction is run in pounds; the dollar figures use $1.34 per £.

ELSEWHERE IN EUROPE

Auctions with no bidders

Denmark

December 2024: **zero bids** for 3 GW in the North Sea. No subsidy, the developer paid the state, and the state took 20%. May 2025: relaunched with 20-year CfDs. August 2026: Vattenfall won two sites at DKK 504 and 542 per MWh.

Germany

June 2023: developers paid €12.6 bn for 7 GW of seabed. August 2025: **zero bids** for 2.5 GW, a first. January 2026: the 2026 tenders were pushed to 2027.

The same lesson in three countries: when the state shifts all the risk to the developer, the developer stays at home.

CHECK · TYPE THE LETTER

Who carries which risk?

1. A merchant farm on a windy, low-price night. **The owner**
2. A CfD farm on the same night. **The state, so in the end the bill-payers**
3. Interest rates rise after the farm is financed at a fixed rate. **Nobody, for that farm; the next one carries it**
4. Interest rates rise between winning the auction and financing the farm. **The developer: Ørsted's Hornsea 4**

BREAK

Five minutes

Back at 19:20. Cameras on when you return.

ØRSTED, 2023–2025

What the arithmetic did to a pure-play

WHEN	WHAT
2023	Ocean Wind 1 and 2 cancelled; DKK 28.4 bn of impairments in nine months; net loss for the year DKK 20.2 bn; dividend paused for 2023–2025
2024	A further DKK 15.6 bn of impairments: US interest rates, seabed lease values, Sunrise Wind
May 2025	**Hornsea 4** (2.4 GW) stopped despite its AR6 contract: supply-chain costs, interest rates, construction risk
Aug 2025	US stop-work order on Revolution Wind, 80% built; later lifted by a court
Oct 2025	A **DKK 60 bn** rights issue; the Danish state holds 50.1%

Ørsted company announcements and annual report 2023.

BUILD, SELL HALF, OPERATE

Ørsted's way to recycle capital

The model

Develop and build a farm, then sell a large stake to an investor who wants long, steady returns (a pension fund, an infrastructure fund). Keep building and operating it for a fee. Use the cash for the next farm.

An example

December 2025: 50% of **Hornsea 3** sold to Apollo for about **DKK 39 bn**. Ørsted remains the builder and the operator. In 2024 it planned about DKK 115 bn of such sales to 2030.

The model works when buyers pay well for steady returns, which means when interest rates are low.

ENI · PLENITUDE

A renewables company inside an oil major

	PLENITUDE
Renewable capacity	just over 4 GW (end 2024) → over 5.8 GW (end 2025) → 6.5 GW expected end 2026
Targets	10 GW by 2028 · 15 GW by 2030
Stakes sold	10% to Energy Infrastructure Partners (≈ €0.8 bn) · 20% to Ares (€2 bn, valuing the equity at ≈ €10 bn)
Today	Eni holds 65%; Plenitude was deconsolidated from Eni's accounts at the end of September 2026
Offshore wind	65% of Vårgrønn (with HitecVision), which holds Plenitude's 20% of Dogger Bank

Eni and Plenitude press releases, 2025–2026. Check the latest quarterly results before citing a capacity figure.

YOUR TWO COMPANIES

Two ways to pay for the transition

	ØRSTED	ENI / PLENITUDE
Where the cash comes from	Selling stakes in finished farms; shareholders (the rights issue)	Oil and gas cash flow; selling stakes in Plenitude itself
Exposed to interest rates	Strongly: large, capital-heavy projects	Less: smaller projects, many technologies, cash from oil and gas
Exposed to oil and gas prices	Hardly	Strongly
Offshore wind	The core of the business	Minority stakes

Neither is safe. They are exposed to different worlds, which is exactly what your Week 4 stress test measures.

NOT ONLY THE DEVELOPERS

The turbine makers lost money too

Siemens Energy

Net loss of **€4.59 bn** in financial year 2023, driven by Siemens Gamesa's quality problems on new onshore turbines.

Vestas

Net loss of **€1.57 bn** in 2022, back to a small profit (about €78 m) in 2023.

Developers squeezed the price of turbines; turbine makers sold below cost; then costs rose. Week 6 picks this up: the supply chain, and China.

EXERCISE 2 · 11 MINUTES · BREAKOUT ROOMS

Bid or walk away?

1. The state offers a **20-year CfD at $122.30/MWh** (the UK AR7 price of £91.20, in dollars).
2. Team 1 is **Ørsted**: cost of capital 7.5%. Team 2 is **a utility backed by an oil major**: 6.0%. The farm: capex $4,560/kW (shallower water, closer to shore), the other inputs as default.
3. On the calculator, find your LCOE. Do you bid? Then: in which of your Week 4 worlds would you **still** bid, and in which would you walk away?
4. Write the decision and one sentence of reasoning in the canvas box "Bid or walk away".

Back at 19:49.

READ-OUT

The same wind, two different answers

A company that pays more for its money needs a higher price for the same wind. Cheaper capital is a competitive advantage in this industry, sometimes a larger one than better turbines.

That is part of the answer to the course question: who should build Europe's wind, and who should own it once it is built.

INTO THE REPORT

Where tonight goes

TONIGHT	REPORT SECTION
Interest rates as a driver; auctions with no bids	**§4 · The bottlenecks**, finance; and a candidate scenario **axis** for §6
Ørsted's numbers, the farm-down model	**§13 · Ørsted** (Phase 3)
Plenitude's numbers, the stake sales	**§14 · Eni** (Phase 3)

The Week 4 page is yours to work through this week: scenarios, the 2×2, the stress test. Phase 1 is due Sunday 15 November.

CLOSE

A wind farm is a bet on the wind, *and on the price of money*

Two thirds of the cost of offshore wind is capital. Whoever gets cheaper money, and longer contracts, builds Europe's wind.

Questions on the material: the forum of the week. Anything personal: the message panel, bottom right of every page.